IMMERSION INTO AFFINE SPACE OR CLOSED IMMERSION INTO A SMOOTH SCHEME OF RELATIVE DIMENSION R

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the observation that both alternatives follow from the displayed factorization; no step is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-immersion-into-affine is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Factorization through affine space of relative dimension r with unramified projection. Same chapter, Lemma lemma-unramfied-over-affine, invoked at proof lines 3552-3553. Supplies the first step of the construction.
- Geometric structure of an unramified morphism as an immersion followed by a standard etale map. Same chapter, Lemma lemma-geometric-structure-unramified, invoked at proof lines 3554-3560. Supplies the factorization with immersions and the standard etale projection.

Source reference. Original lemma 0CBJ: sources/stacks-varieties/pinned-original-source.tex, source lines 3441–3441. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CBK: sources/stacks-varieties/pinned-original-source.tex, source lines 3471–3471. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ with image $s \in S$. Let $V \subset S$ be an affine open neighbourhood of s. If f is locally of finite type and*

$$r = \dim_{\kappa(x)} \Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x) = \dim_{\kappa(x)} \Omega_{X_s/s,x} \otimes_{\mathcal{O}_{X_s,x}} \kappa(x) = \dim_{\kappa(x)} T_{X/S,x}$$

then there exist

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(1) *an affine open $U \subset X$ with $x \in U$ and $f(U) \subset V$ and a factorization*
$$U \xrightarrow{j} \mathbf{A}^{r+1}_V \to V$$
of $f|_U$ such that j is an immersion, or

(2) *an affine open $U \subset X$ with $x \in U$ and $f(U) \subset V$ and a factorization*
$$U \xrightarrow{j} D \to V$$
of $f|_U$ such that j is a closed immersion and $D \to V$ is smooth of relative dimension r.

Proof. Pick any affine open $U \subset X$ with $x \in U$ and $f(U) \subset V$. Apply Lemma 0CBK to $U \to V$ to get $U \to \mathbf{A}^r_V \to V$ as in the statement of that lemma. By Lemma 0CBJ we get a factorization
$$U \xrightarrow{j} D \xrightarrow{j'} \mathbf{A}^{r+1}_V \xrightarrow{p} \mathbf{A}^r_V \to V$$
where j and j' are immersions, p is the projection, and $p \circ j'$ is standard étale. Thus we see in particular that (1) and (2) hold. □